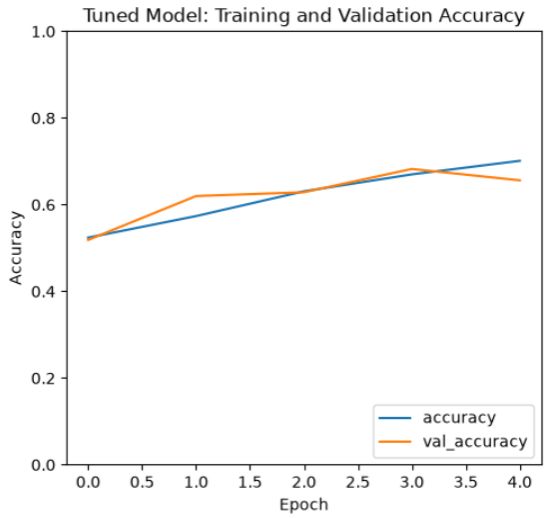
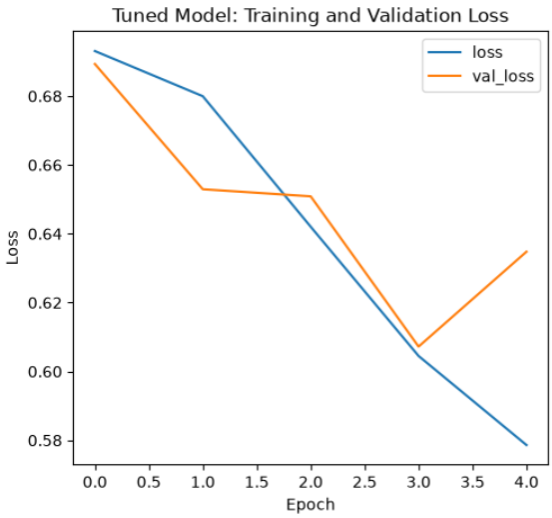
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CNN Activity w/ hyperparameters tuning

Use hyperparameters tuning in keras as discuss in the previous meeting and provide comparison with the manual tuning of hyperparamets.





25/25 2s 55ms/step

25/25 **1s** 27ms/step

Model	Val. Accuracy	Val. AUC
Baseline (manual)	0.6150	0.6542
Tuned (Keras Tuner)	0.6825	0.7373

The tuned model reached 68.25% validation accuracy, compared with 61.50% for the manual baseline, an increase of 6.75 percentage points. Validation AUC rose from 0.6542 to 0.7373, suggesting better separation between the two classes across classification thresholds. The results indicate an improvement for this particular validation split, although classification performance remains modest.

The tuner evaluated only two candidate configurations, and both the baseline and the search trials trained for three epochs. The separate training of the selected configuration for up to five epochs means the comparison does not isolate the effect of hyperparameter choice alone. Because the validation data was used to select the tuned model, these validation

metrics should not be presented as performance on unseen test data. No separate test accuracy or timing measurement is reported here.

Under the computing limits of my local computer, the adjusted notebook completed the comparison and the tuned CNN performed better on the validation set. Its 68.25% accuracy and 0.7373 AUC provide evidence of improvement over this manual baseline, but the limited search and absence of an independent test set restrict broader claims about generalization. A future run with more trials, longer training, and a separate test set could provide a stronger evaluation if sufficient computing resources are available.